ROYAL CITY LABS
PROCESS CONTROL

WEBSITE — DESIGN PREVIEW

The new Royal City Labs website.

A visual summary of the working build — desktop, tablet and phone, in English and French, including the enquiry form and the interactive states.

PREPARED FOR
Royal City Labs

DATE
4 September 2026

CONTENTS
25 pages

CONTENTS

What is in here.

Every image is a screenshot

Nothing here is an illustration. Each image was captured from the production build running in a real browser, at the viewport size printed in the top-right corner of the page.

Three viewports, two languages

The site is documented at **1600 × 1000** (desktop), **900** and **768** (tablet) and **430 × 932** (phone), and it is served as **en-CA** and **fr-CA**.

This is the summary

A full 120-page edition walks every section of every route, and every phone page top to bottom. What is here is the essential: the visual language at a size you can read, plus the entire site in thumbnail on the next page.

Placeholder content is flagged

The logo is a recreation in your brand palette, the portraits are licensed stand-ins, and no project metric has been invented. Everything waiting on your input is on the last page.

Page numbers run along the bottom edge of every sheet.

STRUCTURE

Site map.

PRIMARY ROUTES × 2 LANGUAGES

/en	Home — the scroll narrative, team, method, solutions, industries
/en/solutions	The five engineering disciplines
/en/industries	Ten sectors, each with its own description
/en/approach	The RCL system, and what stays yours afterwards
/en/about	Team, Canadian standards, bilingual delivery, positioning
/en/projects	How every project is documented
/en/contact	The enquiry form

SOLUTION DETAIL ROUTES × 2 LANGUAGES

/en/solutions/control	Industrial Control
/en/solutions/connect	Industrial Connectivity
/en/solutions/engineer	Industrial Engineering
/en/solutions/develop	Industrial Software
/en/solutions/optimize	Advanced Optimization

MACHINE-FACING

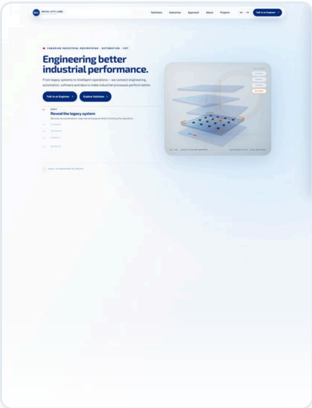
/robots.txt	Crawl rules
/sitemap.xml	All routes, both languages, with hreflang pairs
/	Redirects to /en or /fr from the browser's language

24 page routes in total, every one prerendered as static HTML at build time, so the first paint does not wait on a server.

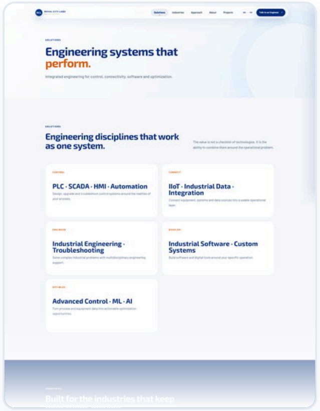
OVERVIEW

The whole site at a glance.

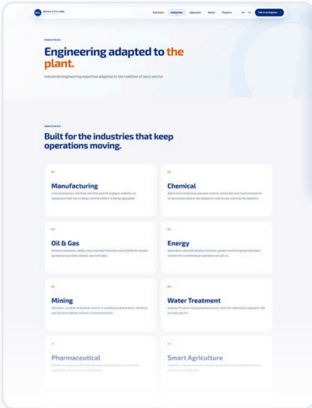
DESKTOP · FULL-PAGE CAPTURES



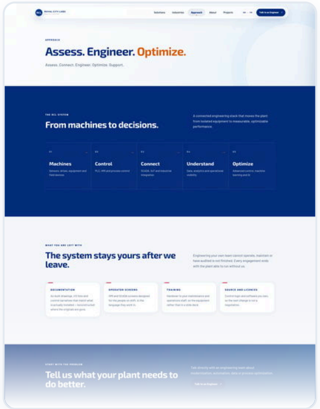
HOME



SOLUTIONS

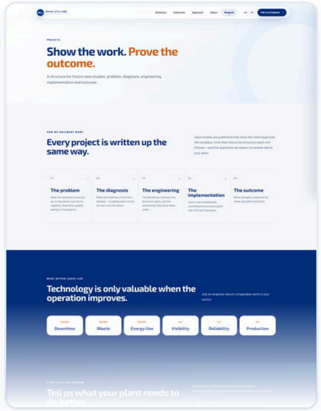


INDUSTRIES

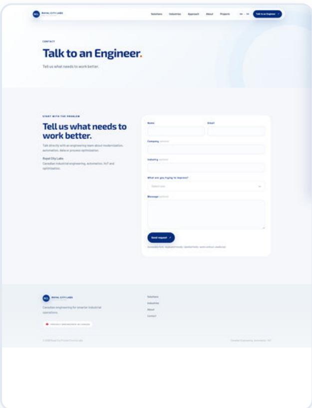


APPROACH

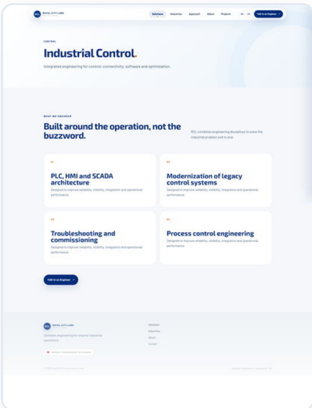
ABOUT



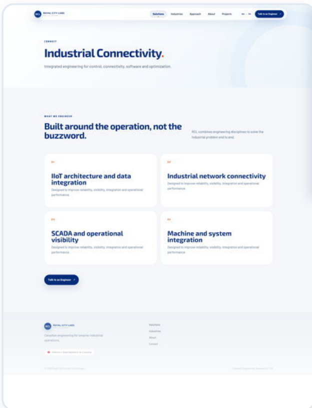
PROJECTS



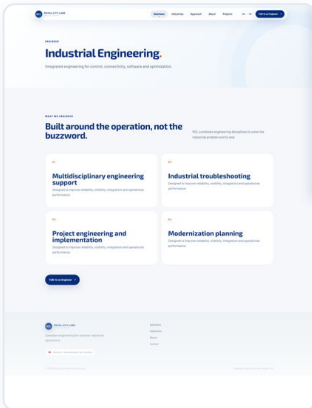
CONTACT



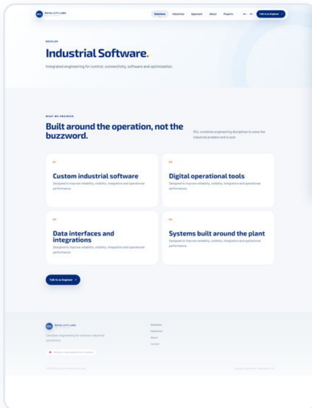
SOLUTIONS › CONTROL



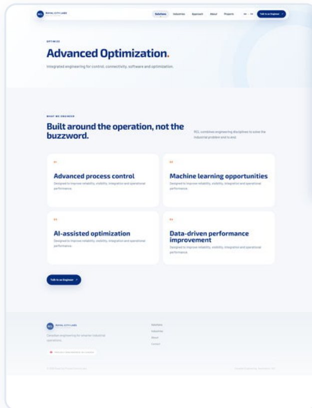
SOLUTIONS › CONNECT



SOLUTIONS › ENGINEER



SOLUTIONS › DEVELOP



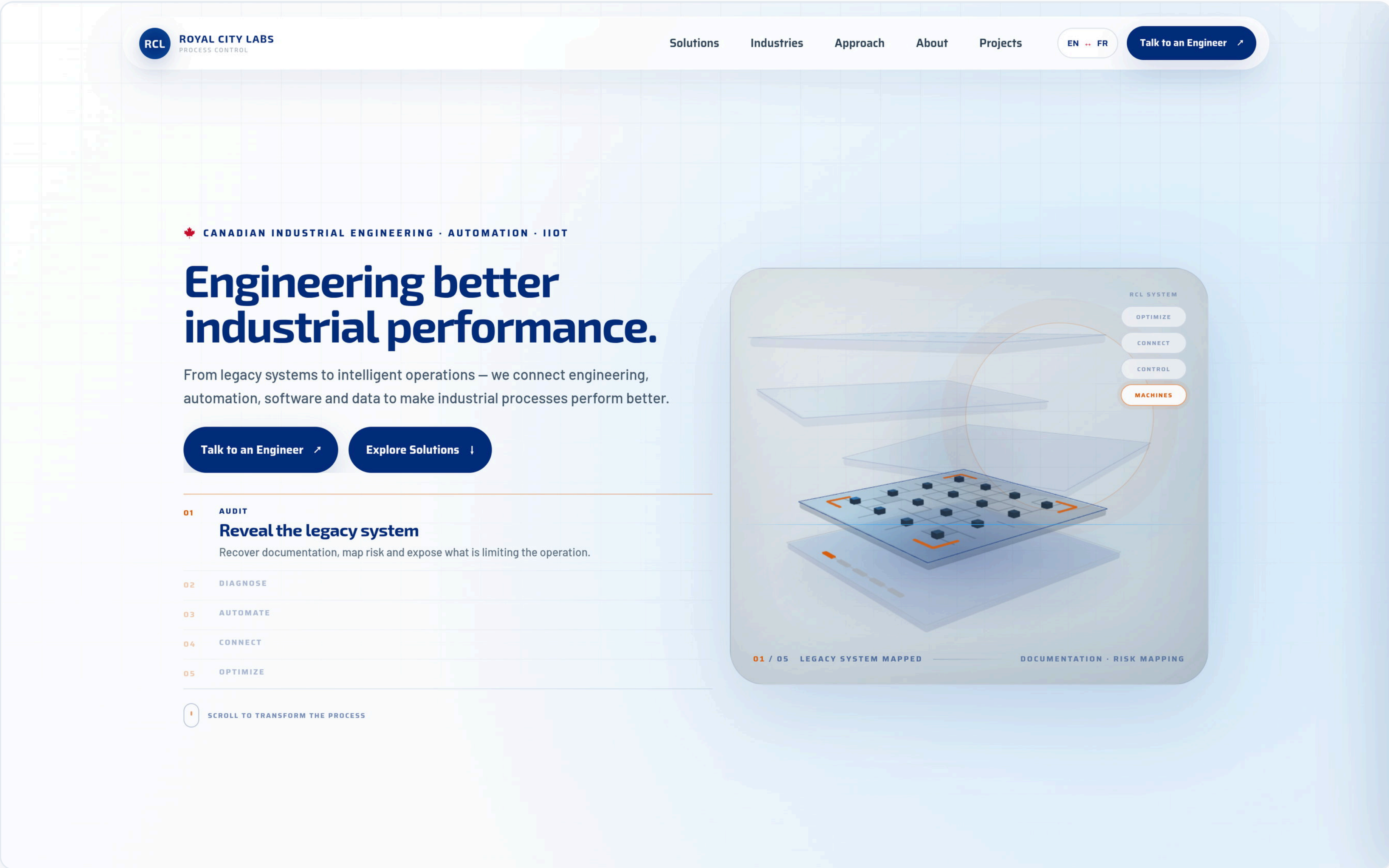
SOLUTIONS › OPTIMIZE

All twelve routes at one scale, each cropped where the sheet runs out. The relative heights show how much there is to read on each. Every one is a live capture.

The first screen

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN



First paint. The headline, both calls to action and the five-step index are in place, and the 3D model shows only the layer the plant already has — three more wait out of register above it. The empty space above the solid layer is the argument: three layers are missing.

HEADLINE + DUAL CTA

FIVE-STEP INDEX

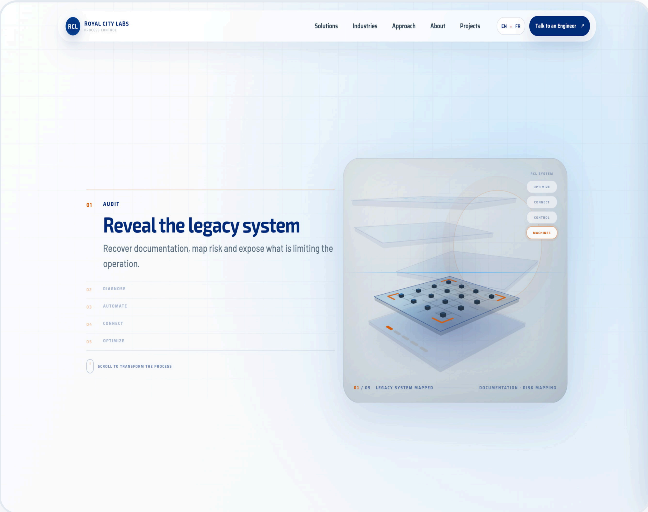
3D MODEL AT REST

02 · HOME

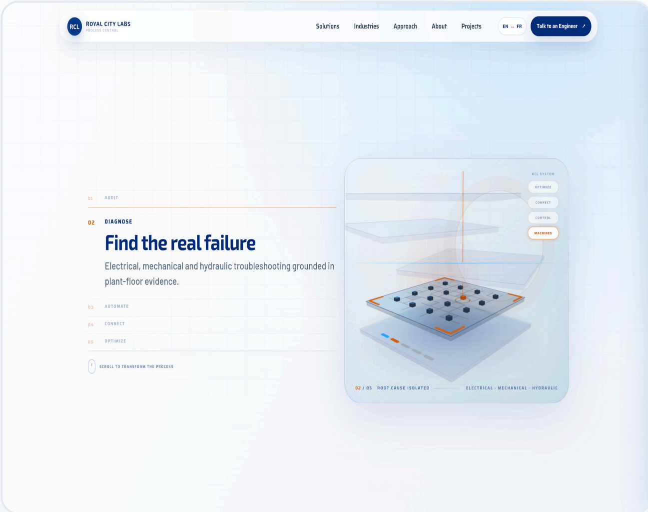
The scroll-driven narrative

DESKTOP · 1600 × 1000

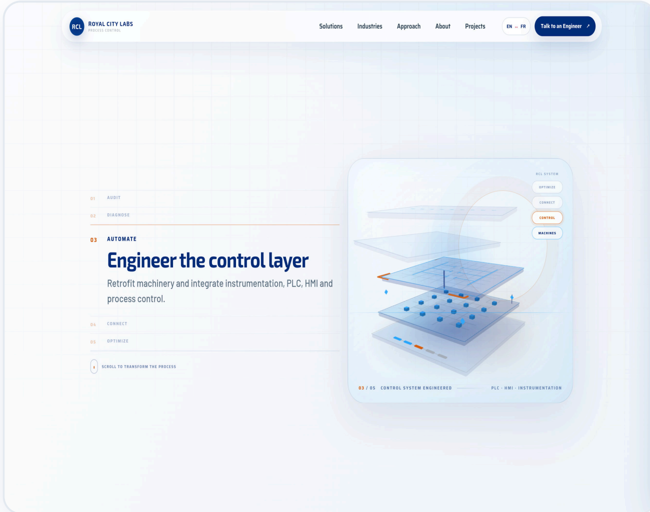
ROYALCITYLABS.CA/EN



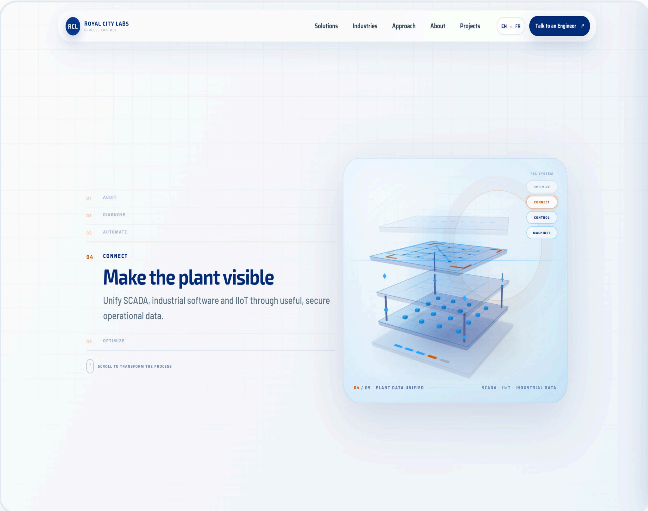
01 · AUDIT



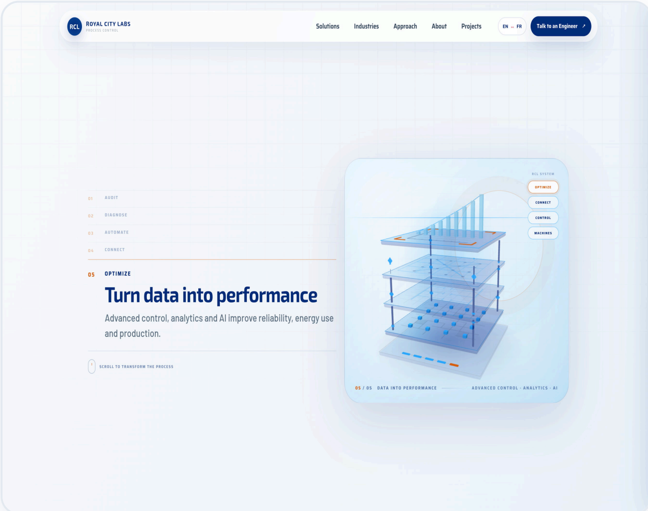
02 · DIAGNOSE



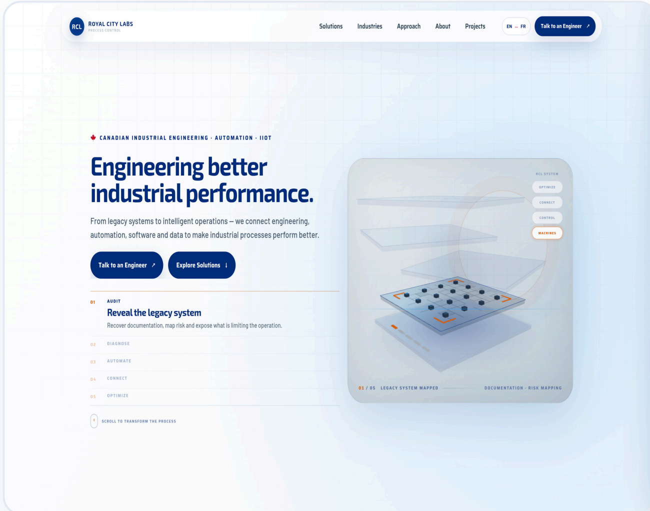
03 · AUTOMATE



04 · CONNECT



05 · OPTIMIZE



AT REST

As the visitor moves down the home page the headline collapses and the five engineering steps open one at a time — while the 3D model assembles itself layer by layer: the plant as it stands, the fault diagnosed, the control layer, the network, and finally analytics with the loop closed. Orange has exactly one job across the site: marking the step being described now.

The engineers, not the vendor

DESKTOP · 1600 × 1000

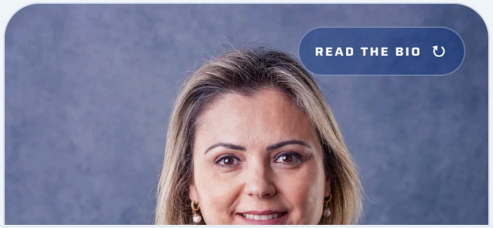
ROYALCITYLABS.CA/EN

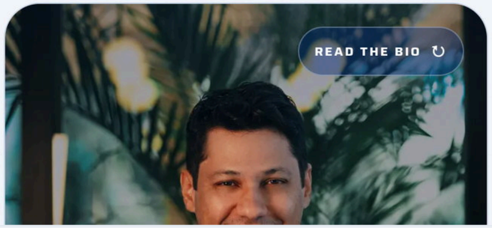
THE PEOPLE BEHIND THE SYSTEMS

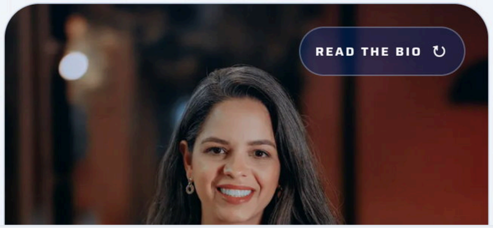
You're not hiring a vendor. You're hiring these engineers.

Growth exposes the plant. The companies that get through it are the ones with senior engineering judgement in the room — not a support ticket. Every RCL project is led by a named engineer with the credentials and the plant hours to own the outcome.

75+	P.Eng.	5	1
YEARS OF COMBINED PLANT EXPERIENCE	CANADIAN-LICENSED ENGINEERS LEADING THE WORK	ENGINEERING DISCIPLINES IN-HOUSE	NAMED LEAD ENGINEER PER PROJECT







A trust strip over flip cards: portrait, tenure, role and credentials on the front; the full biography on the back, on hover or tap. **The figures in the strip are placeholders** pending your confirmation.

The RCL system

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN

THE RCL SYSTEM

From machines to decisions.

A connected engineering stack that moves the plant from isolated equipment to measurable, optimizable performance.

01 →

Machines

Sensors, drives, equipment and field devices

02 →

Control

PLC, HMI and process control

03 →

Connect

SCADA, IIoT and industrial integration

04 →

Understand

Data, analytics and operational visibility

05

Optimize

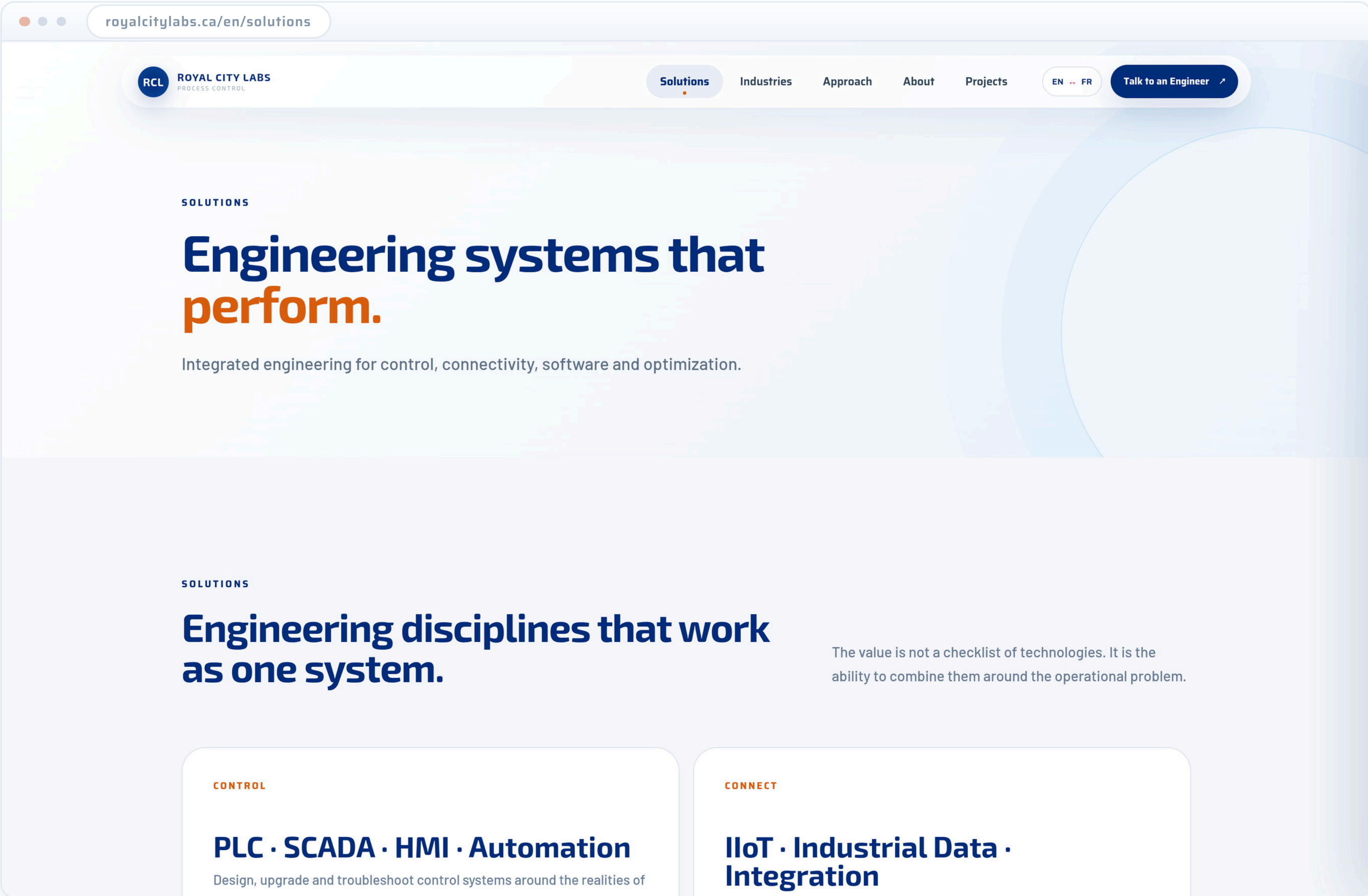
Advanced control, machine learning and AI

Five steps from machines to decisions. This is the same vocabulary the 3D model draws at the top of the page, so the picture and the explanation read as one argument.

Solutions

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN/SOLUTIONS



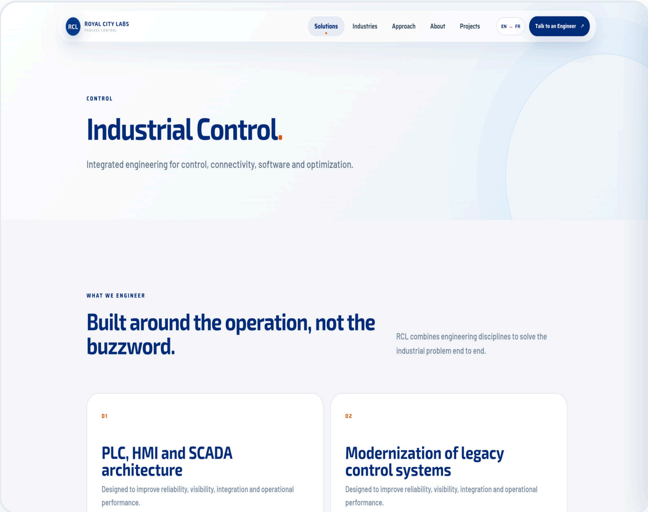
The primary navigation marks the current route with an orange dot under the label, so a visitor who lands here from a search result knows where they are. The page lists the five disciplines and hands each one to a detail route of its own.

03 · SOLUTIONS

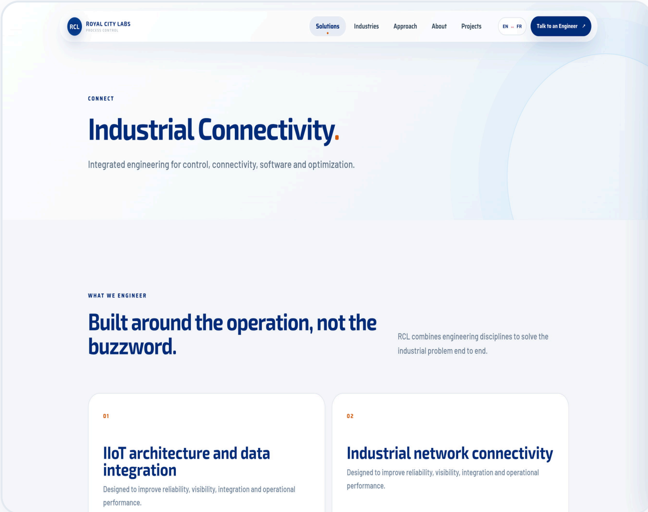
The five detail routes

DESKTOP · 1600 × 1000

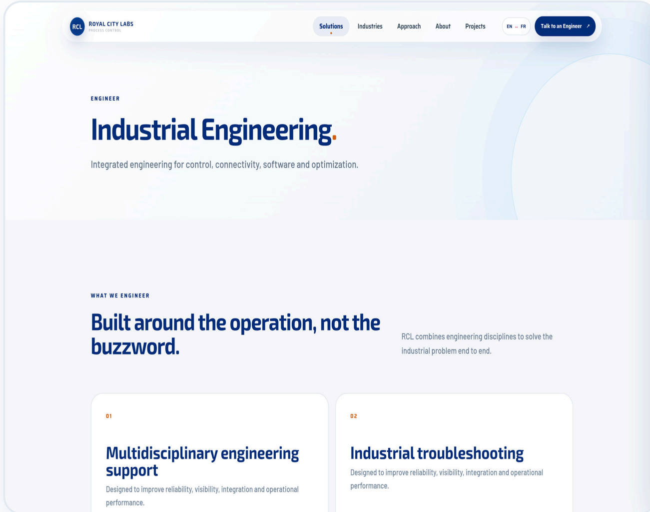
ROYALCITYLABS.CA/EN/SOLUTIONS/…



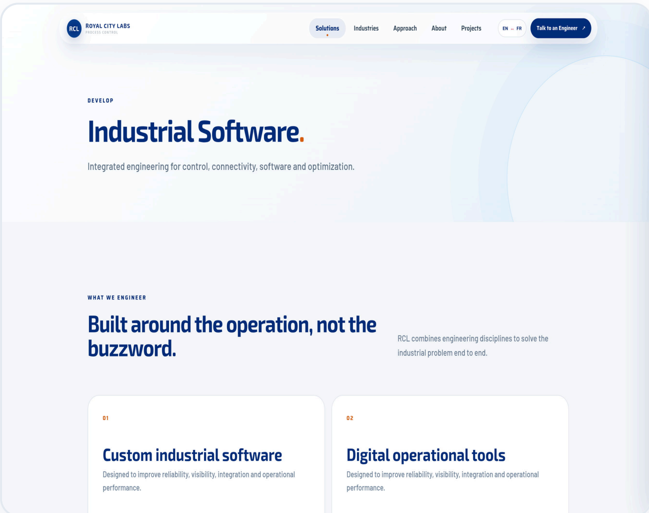
CONTROL



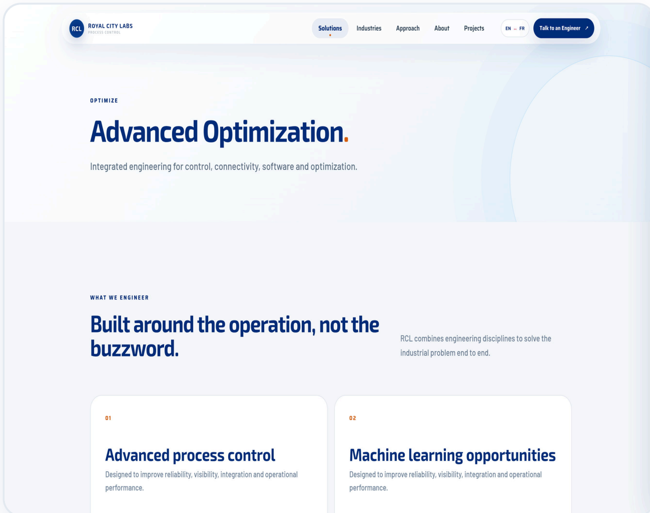
CONNECT



ENGINEER



DEVELOP



OPTIMIZE

Industrial control, industrial connectivity, industrial engineering, industrial software and advanced optimization. Each has its own URL, title tag and meta description, and four capability cards of its own — ten routes in total, counting French.

04 · INDUSTRIES

Ten sectors, ten descriptions

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN/INDUSTRIES

INDUSTRIES

Built for the industries that keep operations moving.

01

Manufacturing

Line automation, machine retrofits and throughput visibility on equipment that has to keep running while it is being upgraded.

02

Chemical

Batch and continuous process control, interlocks and instrumentation for processes where the sequence matters as much as the setpoint.

Each sector carries a distinct description in both languages — all ten are visible in the thumbnail on page 4. The descriptions speak about capability and never claim a result, because no result has been approved for publication yet.

What stays yours

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN/APPROACH

WHAT YOU ARE LEFT WITH

The system stays yours after we leave.

Engineering your own team cannot operate, maintain or have audited is not finished. Every engagement ends with the plant able to run without us.

DOCUMENTATION

As-built drawings, I/O lists and control narratives that match what is actually installed — reconstructed where the originals are gone.

OPERATOR SCREENS

HMI and SCADA screens designed for the people on shift, in the language they work in.

TRAINING

Handover to your maintenance and operations staff, on the equipment rather than in a slide deck.

SOURCE AND LICENCES

Control logic and software you own, so the next change is not a negotiation.

The question an operations director actually asks before signing: as-built documentation, operator screens, training, source code and licences. **This copy is a proposal and needs your review** — it describes how you work.

About

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN/ABOUT

royalcitylabs.ca/en/about

RCL ROYAL CITY LABS
PROCESS CONTROL

Solutions Industries Approach About Projects EN ↔ FR Talk to an Engineer ↗

ABOUT

Built by engineers. Focused on outcomes.

A Canadian engineering and automation company connecting industrial expertise with Industry 4.0.

THE DIFFERENCE IS HUMAN

Multidisciplinary technical talent.

RCL brings multidisciplinary technical talent across engineering, automation, software, data and optimization — so the system is designed around the plant, not around a buzzword.

ROOTED IN CANADA

Canadian engineering for smarter industrial

The credibility route: who the engineers are, the standards the work is held to, the bilingual delivery proof, and the position between traditional industry and Industry 4.0. CSA, IEC and ISO appear as design references, never as certifications.

Bilingual delivery, demonstrated

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN/ABOUT

TWO OFFICIAL LANGUAGES · ONE PLANT

The screen your operator reads is the screen your operator understands.

Canada runs plants in English and in French — often inside the same corporate group, sometimes inside the same building. We author the engineering package in both languages from the start: HMI screens, alarm text, drawings, procedures and floor training. Nothing critical gets discovered in translation after commissioning.

SAME ALARM · TWO LANGUAGES — P-101 · BEARING TEMP

EN Pump P-101 — bearing temperature high

FR Pompe P-101 — température de palier élevée

▲ SAME ALARM · TWO LANGUAGES

HMI & SCADA

Screens, tag descriptions and alarm text written bilingually from day one — not patched in after the plant is running.

DOCUMENTATION

Drawings, functional specifications and O&M manuals issued in the language the site actually works in.

TRAINING

Commissioning support and operator training delivered on site, in English or in French.

Rather than claiming bilingual capability, the page shows it: a sample operator panel with its English and French labels side by side, next to the deliverables that ship in both languages. The difference between saying and proving.

The full team

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN/ABOUT

THE PEOPLE BEHIND THE SYSTEMS

You're not hiring a vendor. You're hiring these engineers.

Growth exposes the plant. The companies that get through it are the ones with senior engineering judgement in the room — not a support ticket. Every RCL project is led by a named engineer with the credentials and the plant hours to own the outcome.

75+
YEARS OF COMBINED PLANT EXPERIENCE

P.Eng.
CANADIAN-LICENSED ENGINEERS LEADING THE WORK

5
ENGINEERING DISCIPLINES IN-HOUSE

1
NAMED LEAD ENGINEER PER PROJECT

Lead — Controls & Automation

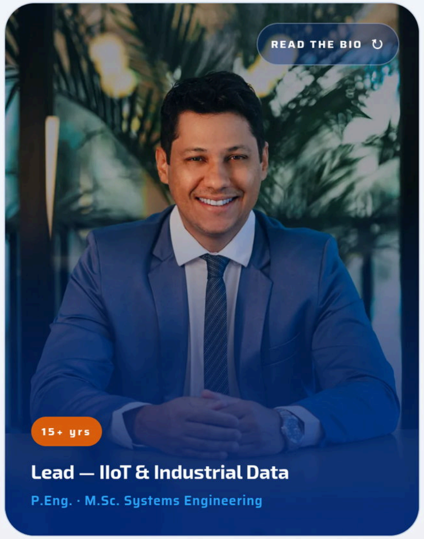
BACK TO PHOTO

P.Eng. · B.Eng. Electrical Engineering

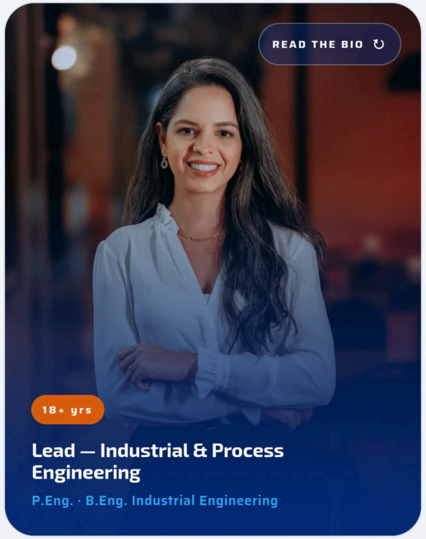
Two decades on plant floors, from single-machine retrofits to multi-line control migrations executed without stopping production. Owns PLC, HMI and safety architecture at RCL — usually the engineer in the room when a legacy system has to keep running while it is being replaced.

PLC · SCADA · Functional safety · Live migrations

LINKEDIN



LINKEDIN



LINKEDIN

The engineer grid with one card turned over, as a visitor would see it while reading. **The portraits are licensed stand-ins** until the real photographs are approved; roles and credentials are proposals for your review.

The documentation framework

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN/PROJECTS

HOW WE DOCUMENT WORK

Every project is written up the same way.

Case studies are published only once the client approves the numbers. Until then this is the structure each one follows — and the questions we expect to answer about your plant.

01 →

The problem

What the operation could not do, in the plant's own terms: capacity, downtime, quality, energy or compliance.

02 →

The diagnosis

What the evidence on the floor showed — including what turned out not to be the cause.

03 →

The engineering

The disciplines involved, the decisions taken, and the constraints they were taken under.

04 →

The implementation

How it was installed and commissioned around a plant that still had to produce.

05

The outcome

What changed, measured the same way before and after.

No case study has been invented. The page presents the five-step structure every project write-up follows — situation, diagnosis, engineering, commissioning, outcome — and says plainly why the numbers are not there yet. Once you approve a real project, it drops straight into this shape.

The enquiry form is not a mock-up

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN/CONTACT

START WITH THE PROBLEM

Tell us what needs to work better.

Talk directly with an engineering team about modernization, automation, data or process optimization.

Royal City Labs
Canadian industrial engineering, automation, IIoT and optimization.

Name

Email

Company optional

Industry optional

What are you trying to improve?

Select one

Message optional

Send request ↗

Accessible form · keyboard friendly · labelled fields · works without JavaScript.

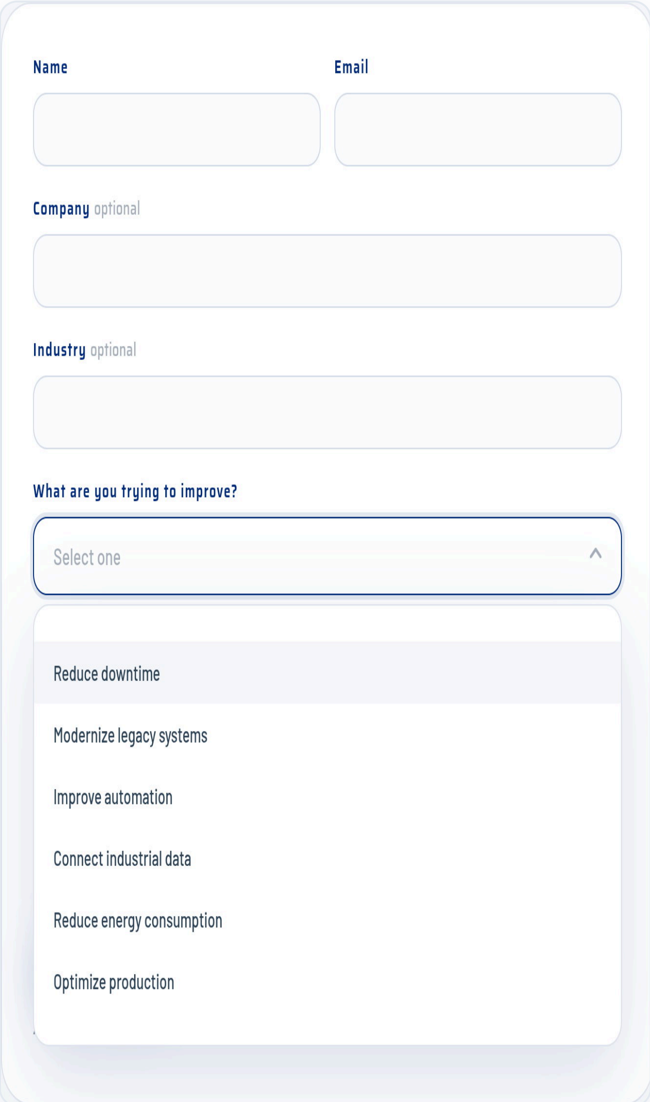
Six fields, two of them required. Labels are permanent — never placeholders that vanish as soon as someone types. Every field is validated on the server, the submission is rate-limited and screened for bots, and it works with JavaScript switched off.

08 · CONTACT

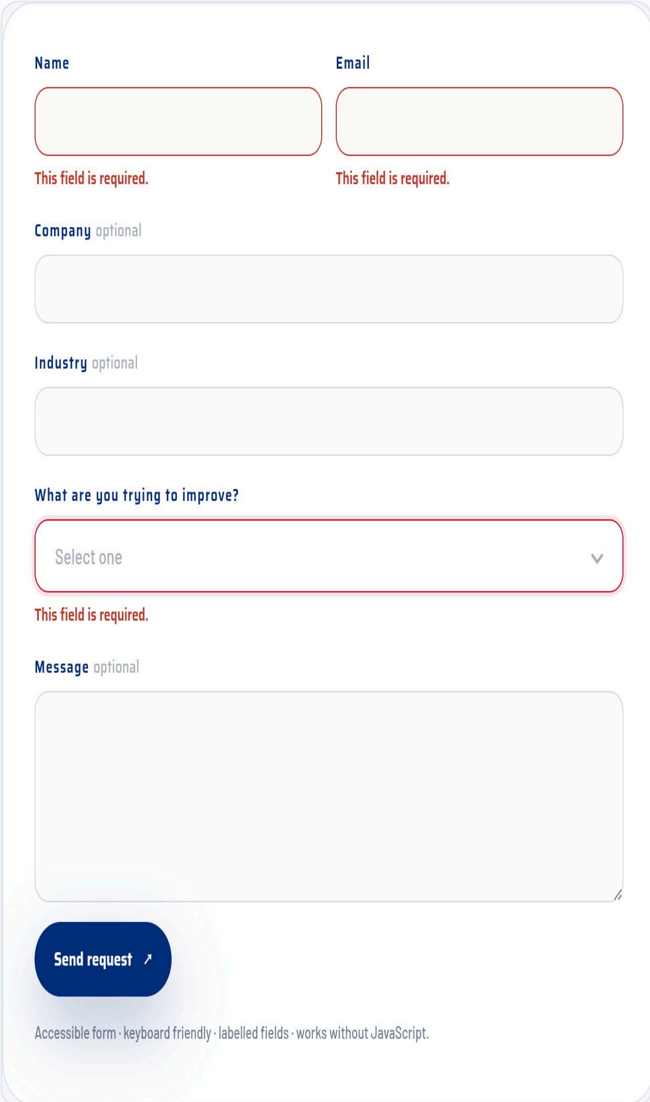
The form in use

DESKTOP · 1600 × 1000

ROYALCITYLABS.CA/EN/CONTACT



SELECTOR OPEN



SERVER-SIDE VALIDATION

CONFIRMATION

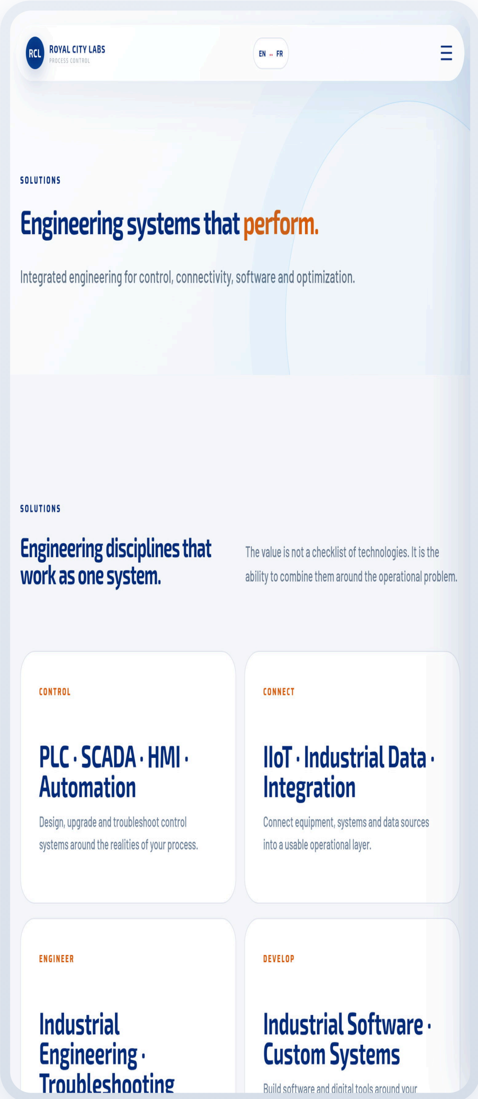
Left: the challenge selector is a custom listbox, keyboard operable, with its values checked against an allow-list on the server. Centre: an empty submission — validation runs on the server, not only in the browser, and each message is announced to screen readers and translated per language; what was already typed survives the error. Right: the confirmation, verified end to end against a live webhook during this review.

09 · TABLET

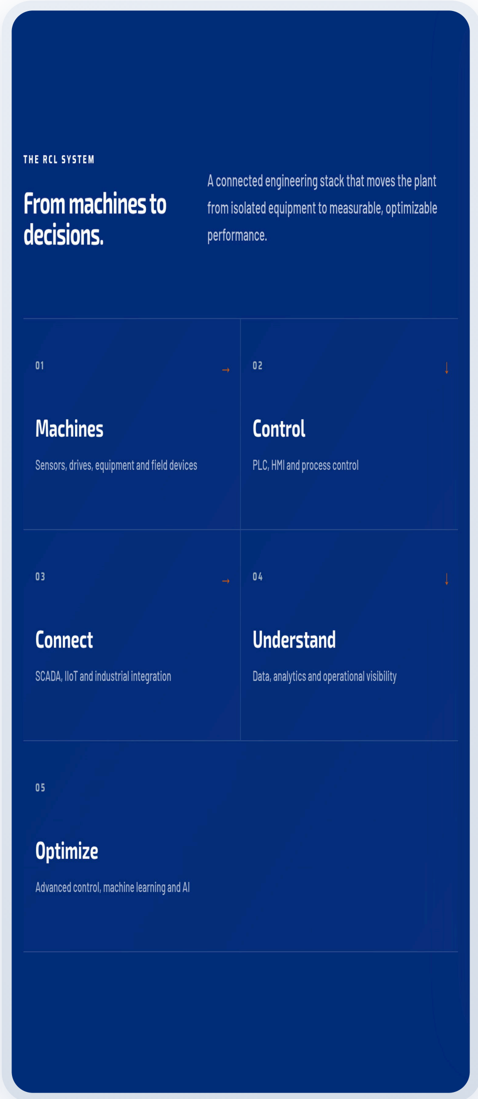
TABLET · 900 AND 768 PX

The layout changes shape on a tablet

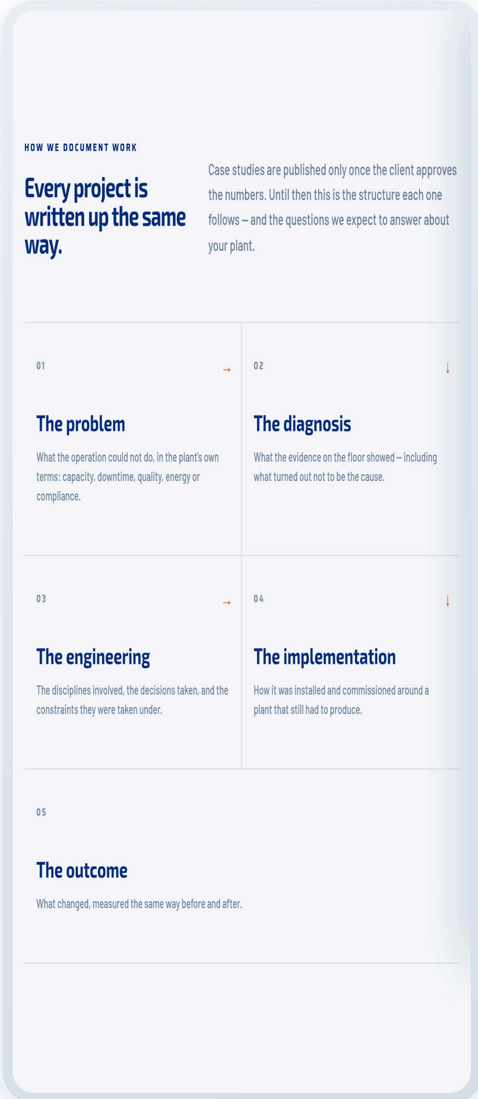
HOME · 900



SOLUTIONS · 900



APPROACH · 768



PROJECTS · 768

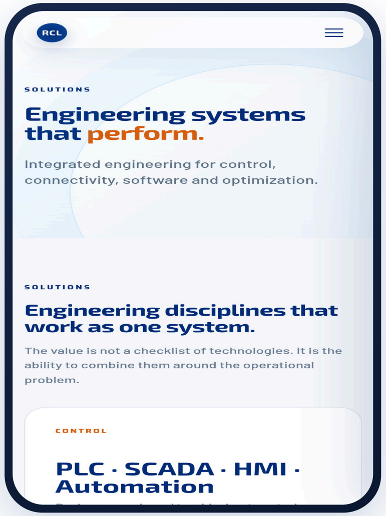
Between 701 and 1100 pixels the primary navigation folds into the menu button, the hero splits, and the multi-column grids reflow. In the two on the right, at 768 px, the five-step grid folds to two columns with the fifth step spanning the width — before this review it was clipped off the edge.

09 · PHONE

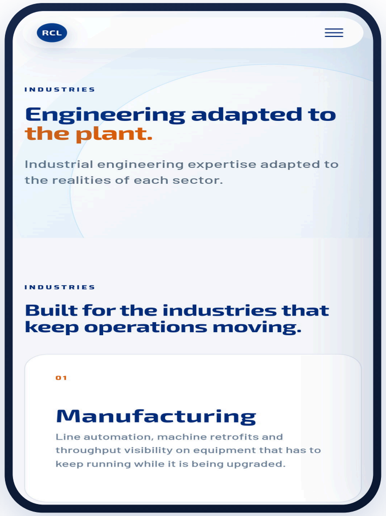
On a phone

PHONE · 430 × 932

HOME

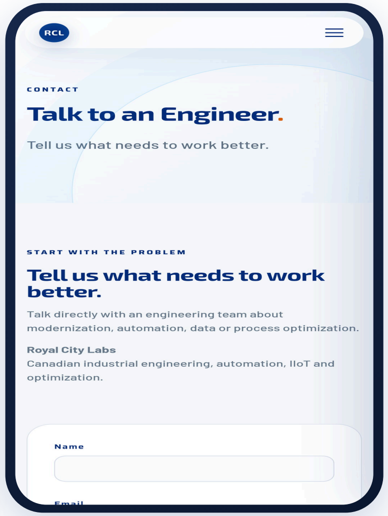


SOLUTIONS

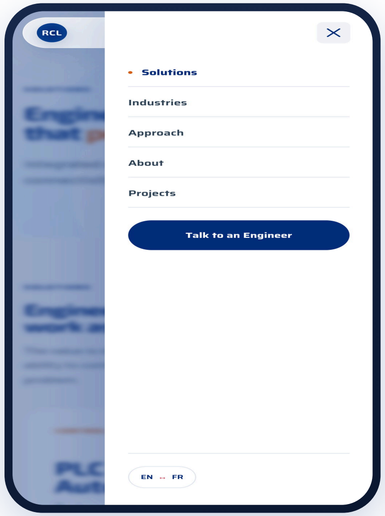


INDUSTRIES

ABOUT

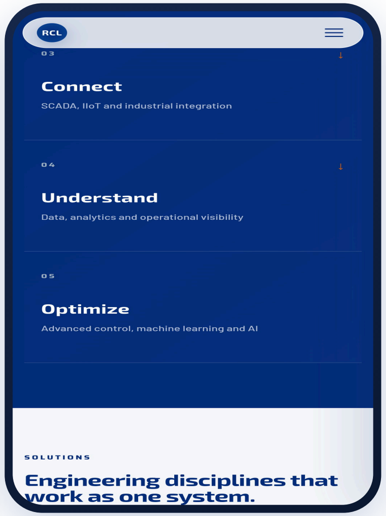


CONTACT

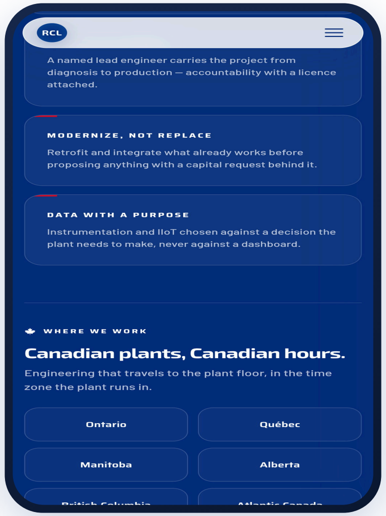


NAVIGATION

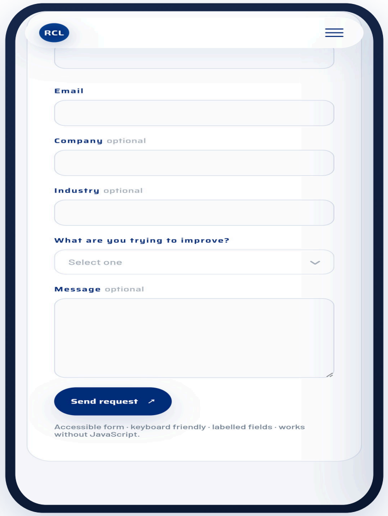
HOME · SCROLL



HOME · SCROLL



ABOUT · SCROLL



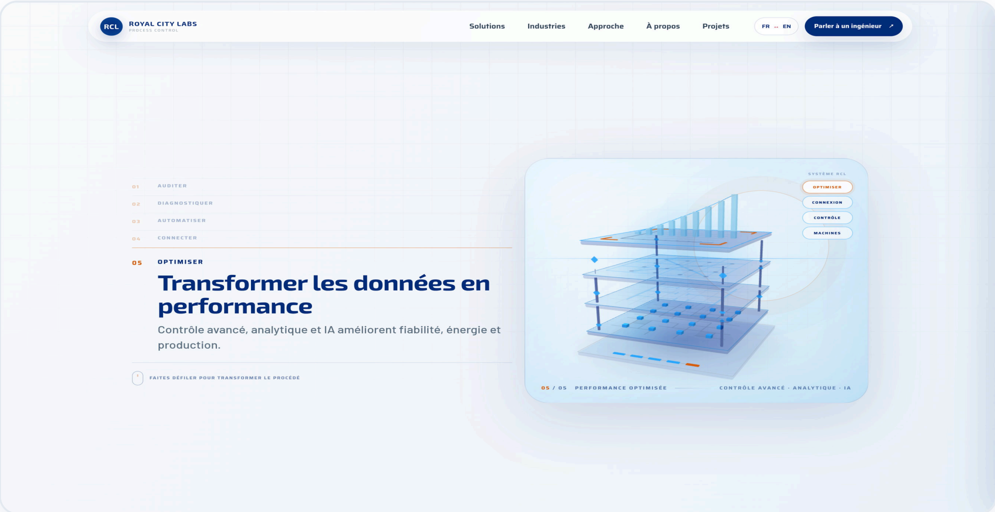
ENQUIRY FORM

Five routes as they load, the navigation panel with the current route marked and the EN ↔ FR switch in its footer, and four screens further down the pages. On a phone the hero presents its finished state rather than animating, so nothing depends on a scroll gesture. The full edition walks all eight routes end to end, frame by frame.

The French site is not a translation layer

DESKTOP · 1600 × 1000

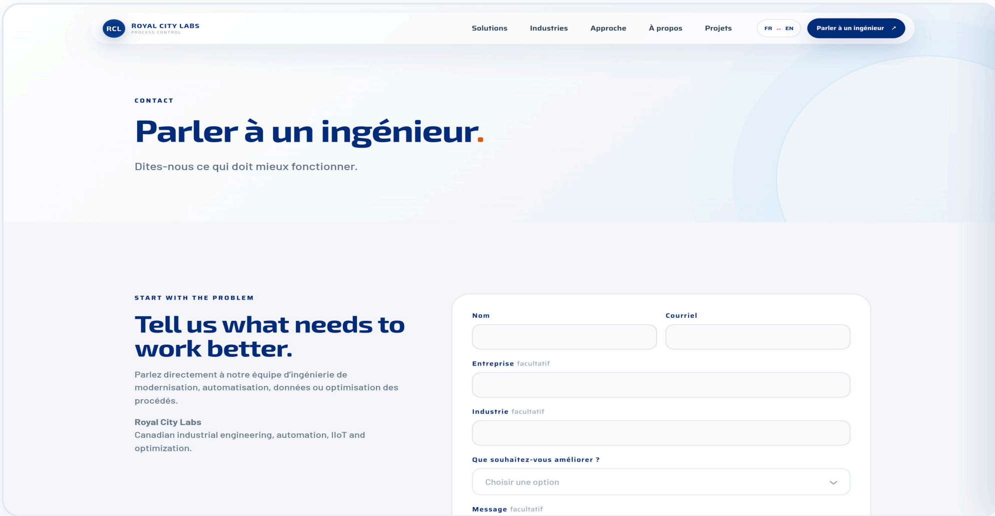
ROYALCITYLABS.CA/FR/…



ACCUEIL

SOLUTIONS

À PROPOS



CONTACT

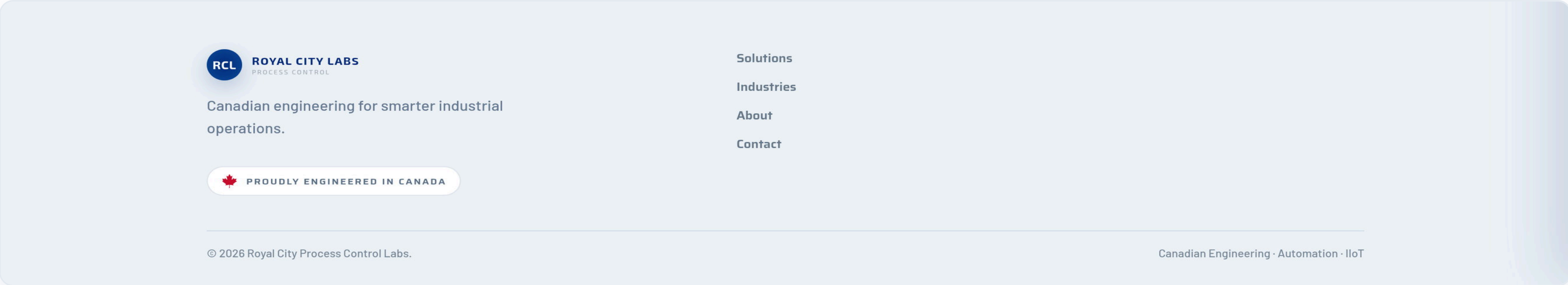
It is twelve routes of its own, each with its own URL, title, meta description and copy, served as fr-CA and paired to the English route by hreflang. Form labels, selector options and every validation message are translated. Switching language keeps the visitor on the same page: /en/industries becomes /fr/industries.

11 · DETAILS

The parts that only exist in motion

DESKTOP · 1600 × 1000

THE NAVIGATION MARKS THE CURRENT ROUTE



FOOTER, ANCHORED ACROSS ROUTE CHANGES

CARD — FRONT

CARD — BACK

The current page is bolder and tinted, with an orange dot under the label — the same orange convention the 3D model uses; screen readers get aria-current="page" on the same link. The footer is anchored, so it does not flicker when a visitor navigates. The engineer card is a real button: it works from the keyboard, and flips to the biography on hover or tap.

TECHNICAL SUMMARY

What it is built on.

Stack

Framework	Next.js 16, App Router
UI	React 19, TypeScript (strict)
3D	React Three Fiber + drei over three.js — geometry generated in code, no multi-megabyte asset
Motion	Framer Motion for reveals; the browser's own View Transitions API for route changes
Styling	Hand-authored CSS on a token system
Type	Exo 2 · Saira · Barlow, self-hosted from your own origin

Delivery and performance

All 24 routes are prerendered as static HTML at build time, so the first paint never waits on a server. Fonts come from your own domain rather than Google's, which removes three round trips before the first glyph. The 3D hero costs around 29 draw calls and fewer than 3,000 triangles, and the render loop stops entirely when the model is off screen.

The enquiry form

- Validated on the server — the browser's own checks are a convenience, not a guard
- Hidden honeypot field plus a rate limit of five submissions per ten minutes per address
- Works with JavaScript disabled
- Delivery is pluggable: a webhook (CRM, Zapier, Make, n8n, Slack) or transactional email

Accessibility

- Correct language on the document: en-CA / fr-CA
- aria-current="page" on the active navigation link, in both navigations
- Form errors wired to their fields with aria-invalid and aria-describedby, and announced
- Visible focus rings on every interactive element
- The whole narrative respects prefers-reduced-motion: the model presents its finished state instead of animating
- The 3D model carries a translated text alternative; decorative overlays are hidden from screen readers

Findability

- Per-page title and meta description on all 24 routes
- hreflang pairs every English route with its French twin
- Organization structured data (JSON-LD) on every page
- robots.txt and a generated sitemap.xml
- / redirects to the visitor's own language

Colour contrast is the one open item: 19 text styles sit below the WCAG AA minimum, the lowest at 1.95:1. All of it is legible — none of it is invisible — and the cause is two brand greys used at small sizes. See the facing page.

QUALITY

What was checked, and what it turned up.

This document was produced by driving the production build in a real browser, which made it possible to test what a code review cannot see. Six genuine faults surfaced — two of them content nobody could read — and all six are fixed and re-verified.

Faults found and fixed

Projects, every width	**The framework descriptions were invisible.** The five-step pattern was written for the dark blue ground it uses on Home and Approach — white body copy, white hairlines — and Projects places it on a light section. White on white.
Seven sections, 14 routes	**Dark-section headings were navy on navy.** The shared heading rule and the dark override carried identical weight, and the shared one was written second — so it won.
About, phones	The bilingual panel sat in a two-column split wider than any phone, cutting off its right edge.
About, phones and tablets	Six region pills stayed six across at every width, pushing "British Columbia" past the edge.
Five-step grids, 701–1100 px	Five columns in a space that fits three clipped step 05 on Approach, Projects and the French home page.
Language switch, phones	A rule that slims the header was also hiding the switch inside the navigation panel — the only one on screen at that width.

Checks that pass

- Linting, TypeScript and the production build: clean
- All 24 routes prerender; no route errors and no runtime errors in the console
- **195 combinations** — 15 pages × 13 viewport widths from 360 px to 1920 px, in both languages — checked for sideways scroll and clipped content: none left
- The 3D narrative renders and advances correctly through all five stages, in both languages
- The enquiry form delivered a lead end to end to a live webhook endpoint
- Every screenshot in this document was taken after the fixes, from the same build

One open item: contrast

An automated sweep of every text style on all 15 pages, in both languages, measured contrast against WCAG AA. **19 styles fall below the minimum** — four of them below 3:1, the lowest at 1.95:1. None of it is invisible and none of it is a bug: it is the brand subtext grey (#A7B0BA) at 7–15 px, a family of muted mid-greys in body copy, and the brand orange at 11 px on white.

Clearing the list means darkening two or three values in the palette — a brand decision, not an engineering one, so we have left it with you rather than quietly changing the look you approved.

Everything else on this page was fixed on sight, because a heading nobody can read is not a matter of taste.

NEXT

What we need from you.

The build is complete and the site works. What is left is content and decisions that are yours to make — nothing on this list needs engineering work, and none of it blocks a decision on the proposal.

Assets

Logo	The official vector. What you see is a careful recreation in your 2026 brand palette — good enough to review, not to ship.
Engineer portraits	Photographs of the real team. The current ones are licensed stand-ins, in place so the card layout could be judged with the photo slot filled.
Favicon	A browser-tab icon and app icons. Currently absent — the browser asks for one and gets nothing.

Content to confirm

Names and credentials	Each engineer's name, role and licence, so the cards can stop showing a role where a name belongs.
The trust figures	Combined plant years, licensed engineers, disciplines in-house. These are placeholders in the code and are marked as such.
LinkedIn	The company page URL, and personal profiles where they exist. The buttons currently all fall back to the company link, which could not be verified.
Proposed copy	The ten industry descriptions and the "what stays yours" list on Approach describe how you work. They are our proposal and need your read.

Decisions

Where leads go	A CRM or automation webhook, or an inbox for transactional email. One setting either way — the pipeline is built and tested.
Domain	The site is built against royalcitylabs.ca; confirm it, and whether French sits on the same domain.
Standards wording	CSA, IEC and ISO are referenced as design practice, never as certifications. If you hold certifications and want them claimed, we need the documentation.
First case studies	Projects deliberately shows the framework instead of invented numbers. Approve one or two projects with real figures and they drop straight in.
Contrast	Darken two or three greys in the palette to clear the accessibility list, or keep the look exactly as it is.

Deliberately not done

No metric, certification or client name has been invented anywhere on the site. Every number you see is either structural — five disciplines, ten sectors, five steps — or flagged in the code as a placeholder awaiting your confirmation. That is why the Projects page has a framework where a competitor would have put percentages.